

Monte Carlo Denoising with a Sparse Auxiliary Feature Encoder

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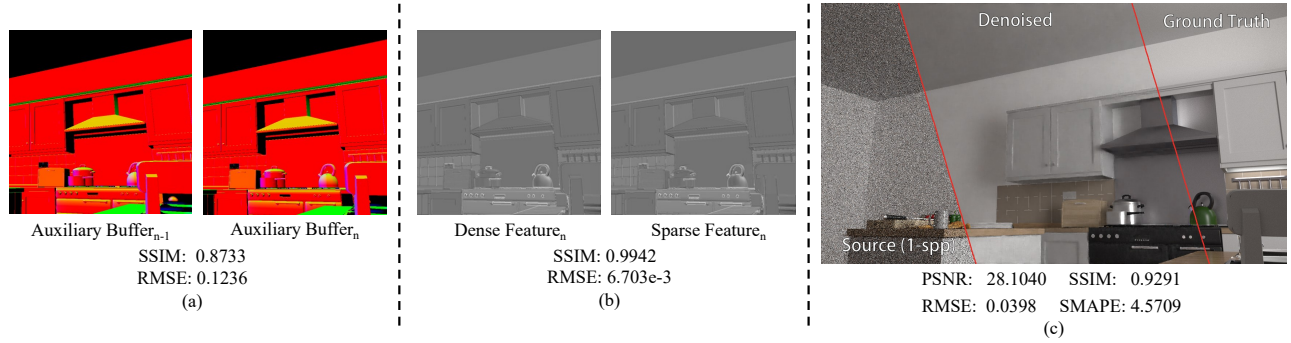


Figure 1: (a) Auxiliary buffers share a strong coherence between frames; (b) Introducing sparse convolutions only slightly changes the features extracted; (c) Our denoising result of the ‘Country Kitchen’ scene, using sparse auxiliary feature encoder.

ABSTRACT

Fast Denoising Monte Carlo path tracing is very desirable. Existing learning-based real-time methods concatenate auxiliary buffers (i.e., albedo, normal, and depth) with noisy colors as input. Such structures cannot effectively extract rich information from auxiliary buffers, however. In this work, we facilitate the U-shape kernel-prediction network with a sparse auxiliary feature encoder. Sparse convolutions can focus solely on regions whose inputs have changed and reuse the history features in other regions. With sparse convolutions, the computational complexity of the auxiliary feature encoder is reduced by 50-70% without apparent performance drops.

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1 INTRODUCTION

Most fast Monte Carlo path tracing algorithms focus on denoising the renderings. Auxiliary buffers, such as albedo, normal, and depth, are the cheap byproducts of the renderer but have rich geometry

and texture information that is essential for denoising. Therefore, neural networks become potential tools for the task, since they can automatically utilize auxiliary buffers without human intervention.

Existing learning-based real-time MC denoising [Fan et al. 2021; Hasselgren et al. 2020; Meng et al. 2020] simply concatenates the auxiliary buffers with noisy images while off-line approaches have adopted independent auxiliary buffer encoders [Lu et al. 2021; Xu et al. 2019]. Also, they only consider the temporal coherence of input and output colors, ignoring auxiliary buffers. To better reuse the history features, we propose a novel sparse auxiliary buffer encoder. Auxiliary buffers between frames are essentially similar (see Fig. 1(a)). Therefore, the encoder can focus solely on the regions whose inputs have changed and reuse the history features in other regions. This meets the core concept of sparse convolutions (see Fig. 3) which has gained attention in video processing [Habibian et al. 2021; Ren et al. 2018].

2 APPROACH

We use a five-stage U-net to generate the multi-resolution kernel filters (E.g. MR-KP [Hasselgren et al. 2020]). At each stage in the decoder, we stack Conditioned Feature Modulations (CFMs) [Xu et al. 2019] after the convolution layers to modulate the feature maps based on auxiliary features. Our auxiliary feature encoder contains four convolution layers and leaky-ReLu functions. The specific structure can be seen in the supplementary material.

Generally, sparse convolutions need masks to choose blocks. A simple form of the pixel-wise mask is the 1-norm of the difference

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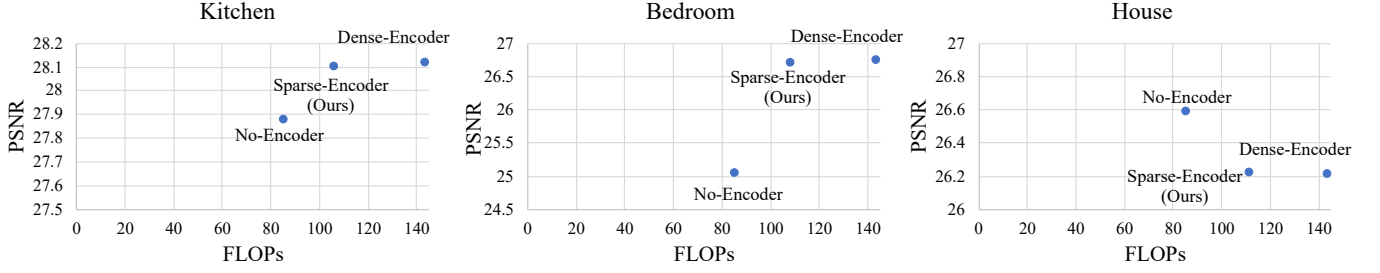


Figure 2: Comparison of average Peak Signal to Noise Ratio (PSNR) and Floating-Point Operations (FLOPs) for the three approaches.

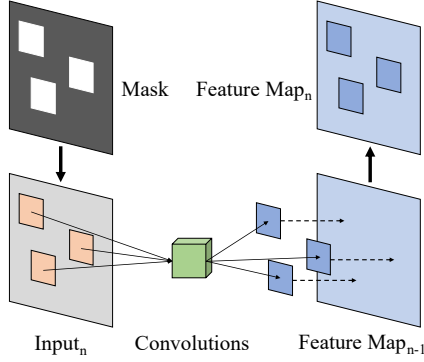


Figure 3: A demonstration of the sparse convolution. The mask will tell the kernel what blocks to take into consideration. After the convolution, the new features will replace the counterparts in the history feature map (FeatureMap_{n-1}) and form the whole feature map (FeatureMap_n). More details of our proposed sparse encoder can be seen in the supplementary material.

between auxiliary buffers, specifically:

$$M_{1 \times 1}(\mathbf{a}_n) = \text{ReLu}(\|\mathbf{a}_n - \mathbf{a}_{n-1}\| - \epsilon) \quad \text{if } n > 1, \quad (1)$$

where $\mathbf{a}_n$ denotes the auxiliary buffer of the n th frame, and ϵ denotes the thresh-hold. With the ReLu function, all elements no more than the thresh-hold will be discarded.

Since typical convolutions' reception fields are larger than a single pixel, the regions around the chosen pixels can't be ignored. Therefore, the pixel-wise mask needs dilation. We dilate the $M_{1 \times 1}$ using a 5×5 filter (all weights are 1). Also, current implementation of sparse convolutions requires block-wise structured masks [Habibian et al. 2021] so we use m times down-sampling and up-sampling in sequence to form an $m \times m$ block-wise mask $M_{m \times m}$:

$$M_{m \times m}(\mathbf{a}_n) = \begin{cases} \text{Step}(\text{Up}_m(\text{Down}_m(F * M_{1 \times 1}(\mathbf{a}_n)))) & \text{if } n > 1 \\ \mathbf{I}^{H, W, 1} & \text{if } n = 1, \end{cases} \quad (2)$$

where F denotes the 5×5 filter, $*$ denotes the convolution, $\mathbf{I}^{H, W, 1}$ denotes the tensor of $H \times W \times 1$ shape with all elements set to be 1, and Step denotes the binary step function which will replace all positives with 1.

3 RESULTS

In the experiment, the block shape of masks is 4×4 and the threshold (ϵ) is 0.5. The training data contain 7 scenes which have 20-100 frames. The testing data contain 3 scenes with 60 frames per each.

With a dense auxiliary feature encoder and CFM, we see improvement in two of three test scenes (see Fig. 2). Introducing sparse convolutions into auxiliary feature encoder slightly changes the performance while the sparsity is 50-70%. Theoretically, the computational costs will decrease by that level since the input and output shape of convolution layers are reduced that much.

The performance of the 'Victoria Style House' scene decreases when using an auxiliary buffer encoder. This is due to the heavy spatial aliasing of its 1-spp auxiliary buffers (see Fig. 2 in supplementary material), which didn't occur in the training data. However, since auxiliary buffers' generation is cheap, using a higher sampling rate or anti-aliasing such as TAA could solve this problem without significant costs (might introduce variation to auxiliary buffers but dealing with which is what neural networks are skilled in).

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REFERENCES

- H. Fan, R. Wang, Y. Huo, and H. Bao. 2021. Real-time Monte Carlo Denoising with Weight Sharing Kernel Prediction Network. *Computer Graphics Forum* 40, 4 (2021), 15–27.
- Amirhossein Habibian, Davide Abati, Taco S. Cohen, and Babak Ehteshami Bejnordi. 2021. Skip-Convolution for Efficient Video Processing. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 2695–2704.
- J. Hasselgren, J. Munkberg, M. Salvi, A. Patney, and A. Lefohn. 2020. Neural Temporal Adaptive Sampling and Denoising. In *Computer Graphics Forum*. 147–155.
- Y. Lu, S. Fu, X. H. Zhang, and N. Xie. 2021. Denoising Monte Carlo renderings via a multi-scale featured dual-residual GAN. *The Visual Computer* (2021).
- Xiaoxu Meng, Quan Zheng, Amitabh Varshney, Gurprit Singh, and Matthias Zwicker. 2020. Real-time Monte Carlo Denoising with the Neural Bilateral Grid. In *Eurographics Symposium on Rendering - DL-only Track*, Carsten Dachsbacher and Matt Pharr (Eds.). The Eurographics Association. <https://doi.org/10.2312/sr.20201133>
- Mengye Ren, Andrei Pokrovsky, Bin Yang, and Raquel Urtasun. 2018. SBNNet: Sparse Blocks Network for Fast Inference. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*.
- Bing Xu, Junfei Zhang, Rui Wang, Kun Xu, Yong-Liang Yang, Chuan Li, and Rui Tang. 2019. Adversarial Monte Carlo denoising with conditioned auxiliary feature modulation. *ACM Trans. Graph.* 38, 6 (2019), 224:1–224:12. <https://doi.org/10.1145/3355089.3356547>